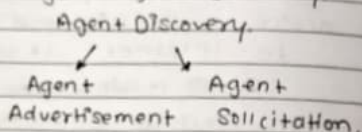


When mobile node is in foreign network then it is necessary to connect it to the Router Foreign Agent. To get this it is important to get agent discovery.



Agent Discovery is the process by which a mobile Node (MN) determines:

- Whether it is a Home network or Foreign network

Agent Solicitation is a message sent by the mobile Node when it does not receive any Agent Advertisement.

Agent solicitation → Mobile node sent Solicitation message to network. All MN will send Solicitation message & it can flood the network. Therefore MN enter in foreign network it can send only 3 message from 3 message 1 msg sent to MN and Foreign Agent sent reply to MN and get connected.

Agent Discovery.

19/28

Agent discovery is the process by which mobile Node (MN) finds out

- Whether it is in its Home network or a foreign network.
- The presence of a Home Agent or Foreign network.
- The care of Address (CoA) it should be use. It is part of Mobile IP protocol.

Mobile Advertisement.

Is a message sent by a Home agent (HA) or Foreign Agent (FA) to inform mobile nodes about their presence in the network.

In Agent Advertisement Agent periodically arrange the advertise with the help of Special Admessage. It is a Beacon broadcast (when signal transmitted diff. diff. types of nodes will get connect).

Q. Explain agent advertisement and discovery registration in mobile networks.

(May 18, 5 Marks)

- When a mobile node is first turned on, it can either be in its home network or a foreign network.
- Hence, the first thing that it must do is to determine where it is, and if it is not at home, must begin the process of setting up datagram forwarding from its home network to the current location.
- This process is accomplished by communicating with a local router serving as an agent (FA), through the process called agent discovery.
- Agent discovery process makes it possible for an MN to determine :
 - o Whether it is connected to its home network or to a foreign network. *mobile network*
 - o Whether it has changed its position. *location*
 - o To obtain a COA when it changes to a different foreign network. *MN/FN*

After moving to another network one initial problem is how to find a foreign agent. For this purpose, mobile IP describes two messages: Agent Advertisement and Agent Solicitation.

3.3.3(a) Agent Advertisement

MU - Dec. 15, Dec. 17

Q. Explain Agent advertisement in Mobile IP.

(Dec. 15, 5 Marks)

Q. How the agent could be discovered using Mobile IP? Give the overlay of agent advertisement packet which includes mobility extension.

(Dec. 17, 10 Marks)

- How does a mobile node make out that it has changed network recently? This is achieved by messages from home agents and foreign agents.
- Home agents and foreign agents advertise their presence and services using messages called **agent advertisement**.
- Agent advertisement messages are periodically broadcast and contain the following details
 - o List of COAs available for the MN.
 - o Special features and services provided by FA such as different types of encapsulation available. For example, minimal encapsulation or generic encapsulation.
 - o Allows MN to detect the network number and congestion details of a link to the Internet.

For agent advertisement ICMP messages with some mobility extension are used, shown in Fig. 3.3.3.

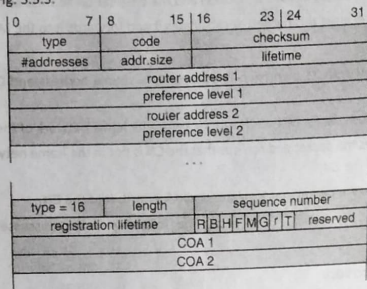


Fig. 3.3.3 : Agent advertisement message

The various fields of ICMP part of the packet are :

- o **Type** : It is set to 9 for ICMP.
- o **Code** : Set to 0, if the agent also routes traffic from non-mobile node. And set to 16 if the agent only routes mobile traffic.
- o **#addresses** : Indicates the number of addresses advertised with this packet. The actual addresses follow shown in Fig. 3.3.3.
- o **Lifetime** : Denotes the length of time for which the advertisement is valid
- o **Preference level** : Preference for each router address is specified. It helps a node to choose the router. The chosen router will act as an FA for the MN.

The various fields of mobility extension part are :

- o **Type** : It is set to 16 for routing only mobile packets.
- o **Length** : Length depends on the number of COAs provided with the message and it is equal to $6 + 4 * (\text{number of addresses})$.
- o **Sequence number** : It indicates the total number of advertisements set since initialization.
- o **Registration lifetime** : Specifies the maximum lifetime in seconds a node can request during registration.

The following bits specify the characteristics of an agent :

- o **R bit** : It is the registration bit and indicates if a registration with this agent is required even when using co-located COA at the MN.
- o **B bit** : This bit is set if the agent is currently too busy to accept new registration.
- o **H bit** : This bit is set if the agent offers services as a home agent.
- o **F bit** : This bit is set if the agent offers services as foreign agent on the link where the advertisement has been sent.
- o **M and G bit** : M and G bits specify the method of encapsulation used for the tunnel. M for minimal encapsulation and G for generic routing encapsulation.
- o **r bit** : This is reserved and set to zero.
- o **T bit** : It indicates that reverse tunnelling is supported by the FA.

3.3.3(b) Agent Solicitation

MU - Dec. 15, Dec. 17

- Q. Explain Agent advertisement in Mobile IP. (Dec. 15, 5 Marks)
- Q. How the agent could be discovered using Mobile IP? Give the overlay of agent advertisement packet which includes mobility extension. (Dec. 17, 10 Marks)

- Agent solicitation messages are sent by MN itself to search an FA in one of the following conditions.
 - o When no agent advertisements are present or
 - o The inter-arrival time of advertisement message is too high or,
 - o An MN has not received a COA by other means.
- To reduce the congestion on the link the MN can send out three solicitations, per second, as soon as it enters a new network.
- Any agent that receives the solicitation message, transmits a single agent advertisement in response. If a node does not receive an answer to its solicitations, it must decrease the rate of solicitations exponentially to avoid flooding the network.
- After these steps of advertisements and solicitations, the MN can now receive COA, either one for an FA or a co-located COA.
- Now the next step is, the MN has to register with the HA if the MN is in a foreign network.

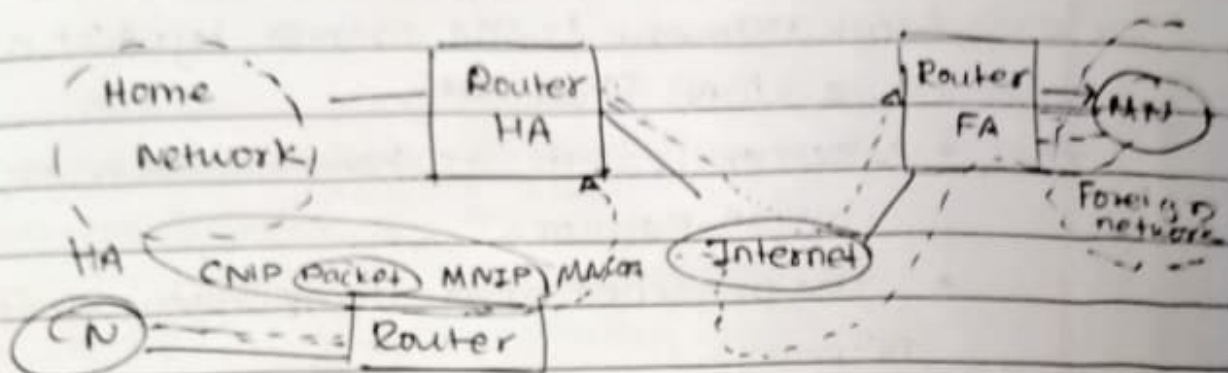
3.3.4 Registration

MU - Dec. 13

- Q. Explain registration with respect to mobile IP. (Dec. 13, 5 Marks)

- After agent discovery is done by a mobile node, it knows whether it is in its home network or in a foreign network. If it is in its home network, it communicates like a regular IP device, but if it has moved to a foreign network, it must activate Mobile IP.
- For activating Mobile IP, a process called home agent registration, or simply registration is used. For registration, the MN exchanges information and instructions with the home agent. The main purpose of registration is to get the Mobile IP working. The mobile node must inform the home that it is on a foreign network so that all datagrams must be forwarded to its foreign network.
- It also must inform the home agent about its care of address (COA) so the home agent can send the forwarded datagrams appropriately.
- When registration is performed, the home agent, in turn, needs to communicate various types of information back to the mobile node.
- Registration can be done in two different ways depending on the location of the COA.
 1. **COA at the FA** : In this case, registration is done as shown in Fig. 3.3.4.
 - The MN sends its registration request to the FA (containing COA).
 - The FA forwards the request to the HA.
 - The HA now setup a mobility binding containing the mobile node's home IP address, the current COA and the lifetime of the registration.
 - The registration expires automatically after the lifetime and is deleted. So the MN should reregister before expiration.
 - After mobility binding, the HA sends reply message back to the FA which forwards it to MN.

Tunneling & Encapsulation



Tunneling is a technique used to forward packets from the Home agent to the Mobile node's care-of address through a virtual tunnel.

It creates a logical path b/w HA & CoA.

Normal Tunneling = HA → MN

Encapsulation.

is the process of wrapping the original IP packet inside another IP packet. It adds a new outer IP header.

Reverse Tunneling

Reverse tunneling is a technique in Mobile IP where packets sent by the Mobile Node are tunneled back to the Home Agent (HA) before being forwarded to the correspondent Node (CN).

Reverse Tunneling = HA → MN
MN → HA

HIPERLAN/2 (High Performance

Radio Local Area Network Type 2) is a wireless LAN standard developed by the European Telecommunications Standard Institute (ETSI).

- Operates in the 5 GHz frequency band.
- Designed to provide high-speed wireless comm.
- Alternative to IEEE 802.11 (Wi-Fi)

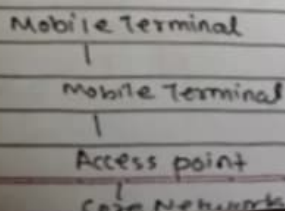
HIPERLAN/2 Architecture

HIPERLAN/2 uses a centralized architecture with the following main components:

- (A) Mobile Terminal (MT)
- User device (laptop, mobile, etc.)
 - Connects to network via Access point.

- (B) Access Point (AP)
- Central Controller of network.
 - Responsible for:
 - Resource allocation
 - Data transmission control
 - Mobility mgmt.

- (C) Core Network (CN)
- provide connectivity to:
 - Internet
 - Other networks (ATM, IP, etc.)



Protocol Stack of HIPERLAN/2

(A) Physical Layer

- Uses OFDM (Orthogonal Freqⁿ Division multiplexing)
- Handles
 - modulation
 - signal transmission

(B) Data Link Control (DLC) Layer Sub-layer

1. MAC (Medium Access Control)
 - Centralized control by AP
 - Time Division Multiple Access (TDMA)
2. Error Control
 - Ensures reliable comm.

(C) Convergence layer

- Connects HIPERLAN/2 with:
 - IP networks
 - ATM networks
- Adapts different network protocols

Adv

- High data rates
- Better Quality of Service
- Strong Security
- Efficient bandwidth utilization

Dis

- Complex architecture
- High cost
- Less popular than Wi-Fi
- Limited commercial adoption

4.3.3 Wireless LAN threats

There are a number of threats that exist to wireless LANS, these include:

- Rogue Access Points/Ad-Hoc Networks
- Evil Twin APs
- Denial of Service
- Configuration Problems also called Mis-Configurations or Incomplete Configurations
- Passive Capturing
- Misbehaving Clients

1. Rogue Access Points/Ad-Hoc Networks

- One way that is the attackers target wireless LANs is by setting up a rogue access point.
- A rogue access point (AP) is a wireless AP that has been installed on a secured network without any authorization from the network administrator.
- The idea is to 'fool' some of the legitimate devices and to make them connect to the rogue access point.

2. Evil Twin Aps

- In this type of attack, the fraudulent AP advertises the same network name (SSID) as a legitimate WLAN, causing nearby Wi-Fi clients to connect to them.
- The only effective defense against Evil Twins is server authentication from 802.1X.

3. Denial of Service

- This is the most common and simplest attack. It can cripple or disable a wireless network by limiting the access to the services.
- This can be done by simply sending a large amount of traffic at a specific target.
- Here the amount of traffic generated to affect a target device is much higher than the capabilities of a target machine.
- A denial of service attack can also be used in conjunction with a rogue access point. For example, a rogue access point could be setup in a channel that is not used by the legitimate access point. Then the denial of service attack could be launched at the channel currently being used causing endpoint devices to try to re-associate onto a different channel which is used by the rogue access point.

4. Configuration Problems

- Simple configuration problems are often the cause of many vulnerabilities. A novice user can set up one of these devices quickly and gain access. However, they also open up their network to external use without further configuration.
- Other issues with configuration include weak passphrases, weak security algorithm deployments (i.e. WEP vs WPA vs WPA2), and default SSID usage.

5. Passive Capturing

- Passive capturing is performed by simply getting within the range of a target wireless LAN and then listening and capturing data.
- This information can be used for a number of things including attempting to break existing security settings and analyzing non-secured traffic.
- It is almost impossible to really prevent this type of attack because of the nature of a wireless network; what can be done is to implement high security standards using complex parameters.

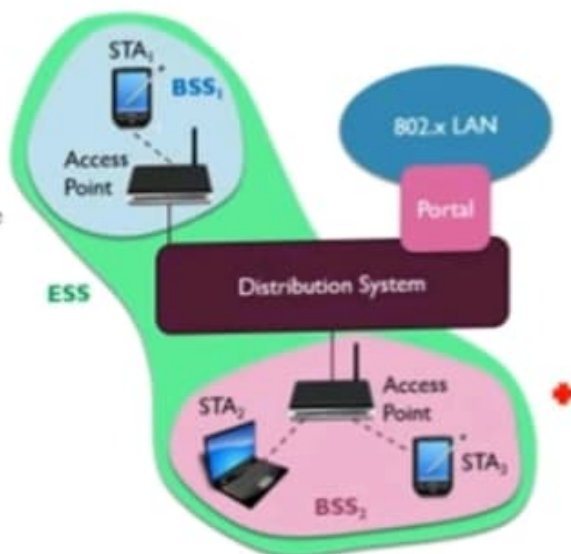
6. Misbehaving Clients

- Sometimes clients form unauthorized Wi-Fi connections accidentally or intentionally. By doing this, they put themselves and corporate data at risk.
- Some enterprises use Group Policy Objects to configure authorized Wi-Fi connections and prevent end-user changes. Others use host-resident agents to monitor Wi-Fi client activity and disconnect high-risk connections.

SYSTEM ARCHITECTURE

Figure shows the architecture

- Several nodes, called stations (STA), are connected to access points (AP).
- Stations are terminals with access mechanisms to the wireless medium and radio contact to the AP.
- The stations and the AP which are within the same radio coverage form a basic service set (BSS).
- The example shows two BSSs – BSS₁ and BSS₂ – which are connected via a distribution system.
- A distribution system connects several BSSs via the AP to form a single network and thereby extends the wireless coverage area.
- This network is now called an extended service set (ESS).
- The distribution system connects the wireless networks via the APs with a portal, which forms the interworking unit to other LANs.



SYSTEM ARCHITECTURE

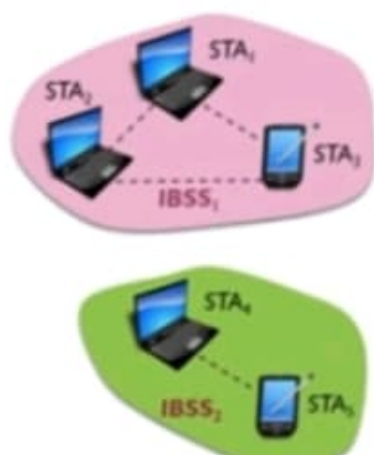
- Every ESS has its own identifier, the ESSID.
- The ESSID is the 'name' of a network and is used to separate different networks.
- Without knowing the ESSID (and assuming no hacking) it should not be possible to participate in the WLAN.
- The architecture of the distribution system is not specified further in IEEE 802.11.
- It could consist of bridged IEEE LANs, wireless links, or any other networks.
- Stations can select an AP and associate with it.
- The APs support roaming (i.e., changing access points), the distribution system handles data transfer between the different APs.
- APs provide synchronization within a BSS, support power management, and can control medium access to support time-bounded service.



SYSTEM ARCHITECTURE

Ad-hoc network architecture

- In addition to infrastructure-based networks, IEEE 802.11 allows the building of ad-hoc networks between stations.
- Every ad-hoc network thus forming one or more independent BSSs (IBSS) as shown in Figure.
- In this case, an IBSS comprises a group of stations using the same radio frequency.
- Stations STA₁, STA₂, and STA₃ are in IBSS₁, STA₄ and STA₅ in IBSS₂.
- This means for example that STA₃ can communicate directly with STA₂ but not with STA₅.



I] Station (STA):

- Several nodes, called Stations (STA), are connected to an Access point.
- All stations are equipped with Wireless Network Interface cards (WNICs) and contain functionality of the 802.11 protocol.
- Wireless station can be mobile devices such as laptop, personal digital assistant, IP phone and other smart phones, or fixed devices such as Desktops and workstations that are equipped with a Wireless Network Interface.

II] Access Point (AP):

- APs are Base station for the Wireless Network.
- APs functions are:
 1. Support roaming
 2. Support power management
 3. Provide synchronization within a BSS

III] Basic Service Set (BSS):

- The stations and access point which are within the same radio coverage form a Basic Service Set (BSS).
- All stations within a BSS communicate with the same access point & compete for shared medium.
- Access point can be connected to other access points via Distribution System.

IV] Extended Service System (ESS):

- A set of connected BSSs together form ESS via AP to form a single network and thereby extend the Wireless Coverage Area.

- This network is now called Extended Service Set (ESS).
- A Distribution System connects every ESS has its own identifier called the ESSID which is 32 byte (max) character string.

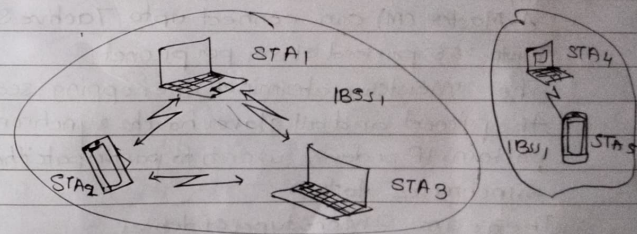
V] Distribution Systems (DS):

- A distribution system connects BSSs via the AP.
- The distribution system connects the wireless networks via APs with a Portal, which forms the interworking unit to other LANs.
- works as a Backbone network & handle data transfer.
- The architecture of the distribution system is not specified further in IEEE 802.11.
- It could consist of bridged IEEE LANs, wireless links, or any other networks.

VI] Portal

- acts as a Internet connecting unit to connect other LANs.

2] Ad-hoc Network Architecture.



- In addition to infrastructure-based networks, IEEE 802.11 allows the building of ad-hoc networks between stations.
- Every ad-hoc network thus forming one or more independent BSSs (IBSS) as shown in fig.

- In this case, an IBSS comprised of a group of stations using the same radio frequency.
- Stations STA₁, STA₂ and STA₃ are in IBSS₁ and STA₄, STA₅ are in IBSS₂.
- This means for example that STA₃ can communicate directly with STA₂ but not with STA₅.

Q) Describe Bluetooth Architecture.

→ There are 2 types of networks made from Bluetooth:

- ① Piconet and
- ② Scatternet.

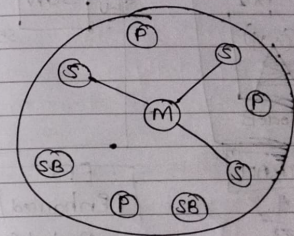
① Piconet : —

- Piconet is a collection of Bluetooth devices which are synchronized to the same hopping sequence.
- Each piconet has only one device called Master (M). All other devices called Slaves (S) are connected to the Master.
- A Master (M) can connect up to 7 active slaves and up to 255 parked slaves per piconet.
- The Master determines the hopping sequence in the piconet and all slaves have to synchronize to this pattern. If a device wants to participate it has to synchronize to this.
- There are 2 more types of devices:
 - i) Parked device (P): cannot actively participate in the piconet but are known & can be reactivated within milliseconds.
 - ii) Standby (SB) devices: do not participate in the piconet.
- All active devices use same hopping seq. & hop together.

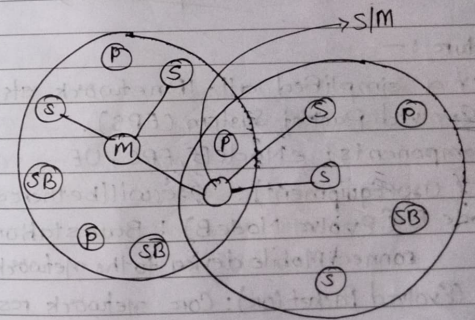
- All active devices are assigned = 3 bit Active Member Address (AMA)
- All parked devices are assigned = 8 bit Parked Member Address (PMA)
- 3 bit AMA
- 8 bit PMA

② Scatternet : —

- Scatternet is a group of piconets. More than one piconet can be connected to form a scatternet through the sharing of a common master or slave device.
- Devices can be slave in one piconet and master in another.
- Communication betn piconets can take place by jumping devices back & forth the piconets.



a) Piconet



b) Scatternet

Discuss in detail about Wifi security protocol:

Wifi-security protocols are standards used to protect wireless networks from unauthorized access, data theft and attacks.

They ensure confidentiality, Integrity and Authentication of data transmitted over wireless networks.

WEP (Wired Equivalent Privacy):

- Introduced : Early Wifi standard
- Security level : Very weak (obsolete)

Features :

- Uses RC4 encryption algorithm
- Same key used repeatedly
- Simple authentication Mechanism

Limitations :

- Easily cracked due to weak encryption.
- Not recommended for modern use.

WPA (Wifi Protected Access):

- Introduced : As an improvement over WEP.
- Security level : Moderate (now outdated).

Features :

- Changes keys frequently.
- Supports Message Integrity check.

Limitations :

- Still vulnerable to certain attacks.
- Not secure enough for modern attacks.

III] WPA2 :

- Introduced : 2004.
- Security level : Strong (widely used)

Features :

- Uses AES (Adv. Encryptⁿ Standard).
- Provides strong encryption and data integrity.

Limitations :

- Vulnerable to KRACK attack if not updated.
- Weak passwords can still be exploited.

IV] WPA3 :

- Introduced : Latest standard.
- Security level : Very strong

Features :

- Provides forward secrecy (past data remains safe even if password is leaked).
- Strong protection against Brute-force attacks.

Limitations :

- Requires newer h/w support.
- Not yet Universally adopted.

Protocol	Security Level	Encryption	Status
WEP	Very weak	RC4	Obsolete
WPA	Moderate	TKIP	Outdated
WPA2	Strong	AES	Widely used
WPA3	Very strong	AES+SAE	Latest

Q7 Explain WLAN and state advantages & disadvantages of W-LAN.

→ Local Area Network: It is Private Network that connects computers, devices within a limited area like building, office, campus, etc.

- Wireless Local Area Network: It is a Wireless Computer network that links 2 or more devices using Wireless (radio) communication to form a LAN within a limited area such as school, building, office, etc.

- The IEEE 802.11 group of standards specify the technologies for wireless LANs.
- Home users can create a wireless network out of an existing wired network and wirelessly extend the reach of the Internet throughout the home on multiple computers.
- By using WLAN, it is now not required that every workstation and conference room be wired up to hubs and switches with cables.

• Advantages:—

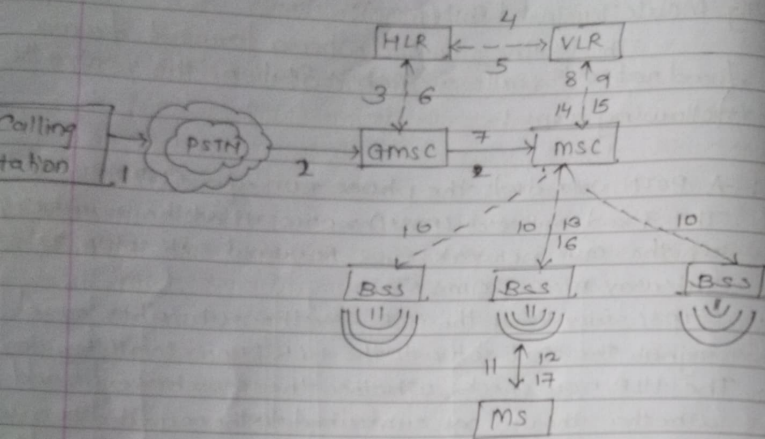
1. Flexibility: Within radio coverage, a user can easily communicate without any restriction.

Simplified planning: Wireless adhoc networks do not require any planning for configuring a network.

3. (Almost) no wiring difficulties: Because no wiring is required, it can be easily installed where wiring is difficult (e.g., historic buildings, firewalls).
4. Robust: Wireless LAN is more robust against disasters like e.g., Earthquake, fire or users pulling a plug.
5. Cost effective: Once a wireless network is installed, adding new additional user will not increase further cost.

• Disadvantages: —

1. Lower Bandwidth and transmission quality: Wireless LAN offers very low bandwidth (1-10 Mbit/s) compared to wired networks due to shared medium. Also it has high error rate due to interference. Hence it offers low QoS as compared to wired network.
2. Many Proprietary Solutions: (data transfer rate)
 $\downarrow$
 $\uparrow$ no device connect $\boxed{\propto}$ rate $\downarrow$ no
3. Local regulatory restrictions: Several countries impose different spectral restrictions. Due to this it is difficult to establish global WLAN solutions.
4. Lower safety & security: Security concerns are high in wireless networks. The open radio interface makes eavesdropping much easier in WLANs than wired network.



Q) What are the characteristics of SIM.

- SIM stands for Subscriber Identity Module and here are its key characteristics in simple terms:

- 1) Unique Identification :- You can each sim have a unique IMSI (International Mobile Subscriber Identity).
- 2) Authentication & Security :- Stores secret keys used to authenticate the user with the mobile networks. Enabled encrypted communication helping prevent fraud and eavesdropping.
- 3) Subscriber Information Storage :-
 - Holds data like:
 - Phone number
 - Network authentication detail
 - Sometimes construct and SMS.

4) Small Standardization forms :-

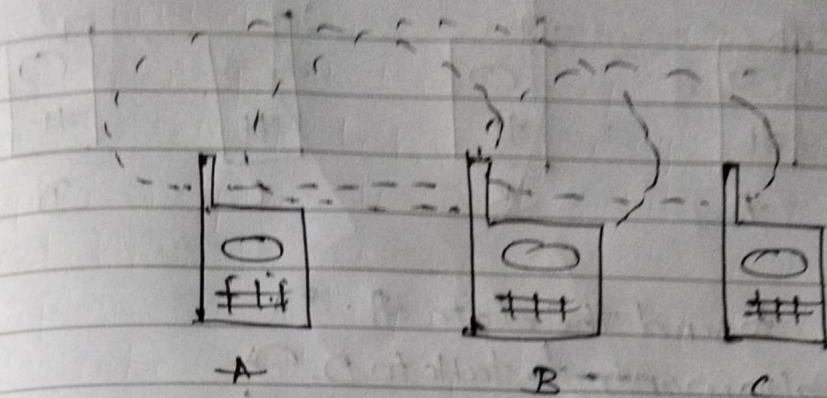
Come in different size standard SIM, SIM, Micro-SIM, Nano SIM, and now eSIM.

5) Operator-Specific :- Issued and managed by mobile network operator, not the phone manufacturer.

Q) Mobile IP: 81screenhat

* Hidden Station & Exposed Station:

Hidden Station:

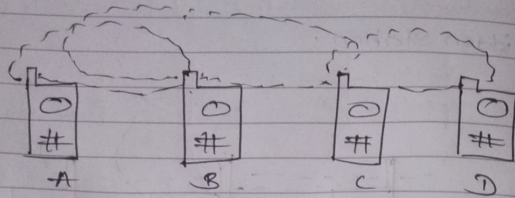


- The transmission range of A reaches B, but not C.
- The transmission range of C reaches B, but not A.
- Suppose the channel is free. B reaches both.
- Suppose Mobile A wants to transmit the data to Mobile B. A senses the channel and finds it free; A transmits to B.
- While A is transmitting, C also wants to transmit to B and hence senses channel.
- But C does not hear A's transmission as it is out of range of C's region.
- So C concludes that Channel B is free and transmits to B.
- Signals from A and C then both collide at B. A is hidden for C.
- This is a Hidden Station problem.

Disadv

- 1) Multiple collisions.

Exposed station:



- B wants to send the data A.
- Now C also wants to talk to D.
- Don't transmit through away from outside the radio range of C. (B is busy).
- Adv.

Under Utilization of Channel.

* How to solve hidden & exposed?

- Hidden & Exposed terminal problems can be solved by using Multiple access with Collision avoidance (MACA) protocol.
- Hidden - Absence of Carrier $\neq$ idle medium.
- Exposed - Presence of Carrier $\neq$ Busy medium.

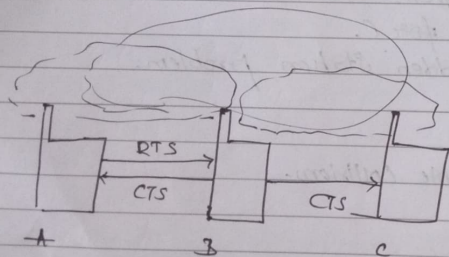


fig. MACA for hidden

- A & C want to communicate to B. A send RTS 1st.

- MACA uses two short signaling packets called RTS and CTS for collision avoidance.

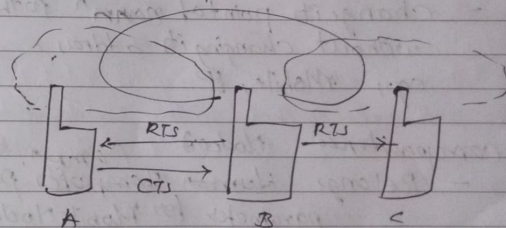
i) RTS (req. to send): A sender req. the 'Right to send' from a receiver by transmitting RTS packet before data transmission.

ii) CTS (Clear to send): Receiver grants 'Right to send' as soon as it is ready to receive by sending back a CTS packet.

- These packets contain sender address, receiver address, length of future transmission.

- Still there are chances of collisions during sending of RTS as both sender could send RTS at same time.
- RTS is very small packet as compared to data packet, so the probability of collisions is much lower.
- In such cases B resolves this contention and only sends CTS to one station.
- MACA solves the hidden prob.

- 1

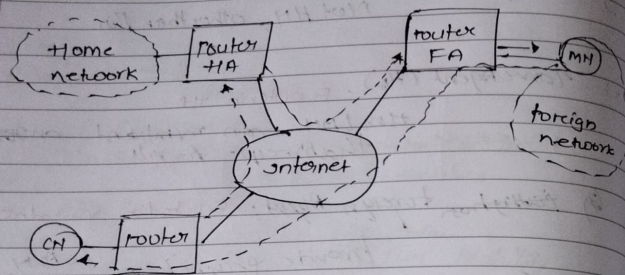


- B wants to send data to A.
- C also wants to send data to someone else (not A/B).
- C does not react to RTS as it is not the receiver.
- A responds to RTS by sending RTS.
- CTS not received by C - (not in range).

- 1871 to mobile mode at 1874

3. Mobile Node received packet directly.

* IP Packet delivery: —



Two Scenarios of data transmission between CM & MN:

- i) CM is fixed node and data is transferred from CM to MN.
- ii) CM is fixed node and data is transferred from MN to CM.

i) Explain Scenario I —

If your Mobile is Home Network and other locⁿ is Foreign Network for your Mobile device.

CM will send packet to the Router & to the Internet. It will go to the Router HA & it will know that MN is not in Mobile (Home Network).

Home agent (HA) will travel to router FA via tunnel and packet encapsulated to FA and packet will be HA CNIP (Packet) MNIP MN's COA.

Now this encapsulated packet will send to Router HA to the FA and get MN locⁿ FA decapsulate the packet & sends message to the MN.

ii) CM is fixed Node (FN) & data is transferred from mobile CM.

MN will send packet to Router FA it will send via Internet, Router to CM with packet & IP addressing.

* Difference: —

Parameters	Adhoc Network (Infrastructure less netw.)	Infrastructure Network (Infrastructure netw.)
Structure	Decentralized, Peer-to-Peer	Centralized logic cannot be found.
Central Node	More access point req'd	Requires access point
Setup	Quick, temporary	fixed, permanent.
Range	Limited to proximity	Extended by accept.
Common use	file sharing, Bluetooth,	WiFi, Internet acc.
	Limited Scalability	Highly Scalable.
	Nodes consume more power	Nodes consume less power
	Lower cost	Higher cost
	NO planning	More planning

7/5/26

MODULE 6

Long Term Evolution of 3GPP.

- ✓ 1) Write a short note on: LTE v12
- ✓ 2) Explain in short voice over LTE. May '23
- ✓ 3) Explain different components used in LTE architecture with diagram. May '23
- ✓ 4) Compare LTE and LTE adv. May '24, MUSA
- ✓ 5) What do you mean by Self Organizing Network. Explain the architecture of SON. May '24, MUSA

* Overview :

GSM, GPRS → 2G

UMTS → 3G

LTE, VoLTE, SD-WAN → 4G

↳ dynamic configⁿ of LTE based stations to automate
↳ transmits voice over LTE network
↳ Higher data rate, low latency.

- 3GPP stands for Third Generation Partnership Project.
- Under 3GPP, widely used 3G standards UMTS WCDMA/HSPA were developed.
- LTE (Long term Evolution) is the 4G successor to 3G UMTS system.
- Came into market around 2010.
- Initially deployments gave little improvement over 3G HSPA & were sometimes treated as 3.5G or 3.9G.
- Later full capability of LTE was realized.
- The first deployments were simply known as LTE, but later deployments were designated 4G LTE Advanced.

Technology	WCDMA	HSPA	HSPA+	LTE	LTE-A
Release	Rel 99/4	Rel 5, 6	Rel 7	Rel 8	Rel 9, 10
Rollout	2003/4	2005-8	2009	2010	2014
Downlink Rate	384 kbps	14 kbps	28 Mbps	150 Mbps	1 Gbps
Uplink Rate	128 kbps	5.7 kbps	11 Mbps	75 Mbps	500 Mbps
Delay	150ms	100ms	50ms	10ms	<10

* Q. LTE: —

- Long Term Evolution (LTE) is a 4G wireless broadband technology development to improve mobile data speed capacity and reliability compared to 3G network.
- It was standardized by 3GPP (Third Generation Partnership Project).
- Came into market around in 2010

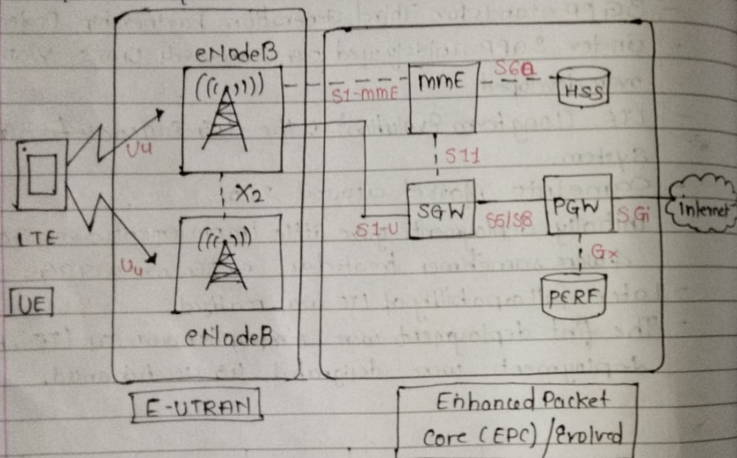


Fig: SAE/LTE Architecture

- LTE is a 4G Wireless Commⁿ technology designed to provide high-speed data, low latency, and improve network performance compared to 3G systems.

- Key features of LTE:

- High data speed
- Download up to 100 Mbps
- Low latency (~10ms)
- Uploaded up to 50 Mbps
- Supports IP-based commⁿ
- Uses SC-FDMA for uplink & OFDMA for downlink

I. SAE Requirements for LTE: —

- Soon it was realized that the system architecture needs to be also evolved in order to support very high data rate and low latency requirement for 3G & LTE.
- Requirements of SAE architecture for LTE include:
 1. Flat arch: consisting of just one type of node, the base station
 2. Effective protocols for support of Packet-switched services
 3. Open interfaces & support of Multivendor equipment interoperability.
 4. Efficient mechanisms for operatⁿ & mechanisms
 5. Support of easy deployment and configⁿ

II. LTE /SAE Architecture: —

- System Architecture Evolution (SAE) is a new network architecture designed to simplify LTE network. It establishes a flat architecture similar to other IP-based commⁿ networks.
- SAE uses eNB and Access Gateway (AGW) and removes the RNC and SGSN from the equivalent 3G network architecture.
- This allows the network to be built with an "All-IP" based network arch. called Evolved Packet System (EPS).
- The High-level network architecture of LTE is comprised of the following three main components:
 - i) The User Equipment (UE)
 - ii) The Evolved UMTS Terrestrial Radio Access Network (E-UTRAN)
 - iii) The Evolved Packet Core (EPC).
- i) User Equipment (UE):
 - User Equipment (UE) is nothing but Mobile Equipment.
 - The Mobile Equipment comprised of the following imp. modules:

- MTC (Machine Type Communication) - handles all commⁿ funcⁿ.
- TE (Terminal Equip.) - Terminates data stream
- UICC (Universal Integrated Circuit Card) - SIM card for LTE Equipment. runs USIM.

ii) E-UTRAN:

- Between E-UTRAN and E-UTRAN there is "Uu interface".
- E-UTRAN handles the radio commⁿ betⁿ the User Equipment (UE) & Evolved Packet Core (EPC).
- Contains only one component, the evolved base station, called eNodeB or eNB.
- Two nearby base station can be connected via the "X2 interface", which is mainly used for signaling & packet forwarding during handover.
- E-UTRAN is connected to EPC via "S1 interface".

iii) EPC:

- Core network responsible for data routing, authentication, mobility mgmt.
- EPC provides means to communicate with packet data networks in the outside world such as Internet, private corporate networks or the IP multimedia subsystem.
- EPC is connected to outside world via "S-GW interface".
- EPC contains foll^y components:
 - a) Serving Gateway (S-GW)
 - b) PDN GW
 - c) Home Subscriber Server (HSS)
 - d) Mobility mgmt Entity (MME).

a) Serving GW (S-GW) -

The Serving gateway (S-GW) acts as a router, and forwards data betⁿ base station and the PDN gateway.

b) PDN-GW -

PDN GW is the point to interconnect betⁿ EPC and External IP networks.

funcⁿ:

- Packet filtering
- Packet screening, etc.

c) HSS -

HSS (Home Subscriber Server) is a Database that contains user-related and subscriber-related info.

d) MME -

MME (Mobility mgmt Entity) handles signaling related to mobility & security for E-UTRAN access.

III) Technical Feature:

- Bandwidth flexibility - supports 104 KHz to 20 MHz channel bandwidth.
- Peak speed -
 - ★ Downloads upto 100 Mbps.
 - ★ Uploads upto 50 Mbps.
- Low latency - Around 10 ms, much lower than 3G.
- Duplex Mode -
 - ★ FDD (Frequency division duplex)
 - ★ TDD (Time Division Duplex).
- Advanced Antenna System - MIMO technology increased data rates and signal quality.

IV] LTE Advanced :-

- Speed upto 1 Gbps.
- Carrier Aggregation.
- Enhanced MIMO support.
- Better coverage and capacity.

LTE Adv meets the requirement of the 4G standard defined by the ITU.

V] Advantages :-

- 1) Higher data speed.
- 2) Better voice quality with VoLTE.
- 3) Improved coverage
- 4) Efficient spectrum usage
- 5) Lower operating cost for service providers.

* Voice over LTE (VoLTE) :-

- Voice Over Long Term Evolution is what VoLTE stands for.
- It is a digital packet voice service that is delivered over IP via an LTE access network.
- LTE is all IP network and the ability to carry voice was not given much importance. Therefore, LTE networks to carry traditional circuit switched voice calls, a different solution way required.
- This solution to carry voice over IP in LTE networks is commonly known as "VoLTE".
- Basically VoLTE systems convert voice ~~data~~ into data stream.

- Voice over LTE, VoLTE is based on IMS (IP Multimedia System) based technique.
- To make implementation of VoLTE easy and cost effective to operators, cut down version of IMS network was defined.
- This not only reduced the no. of entities reqd in the IMS network but it also simplified the interconnectivity.

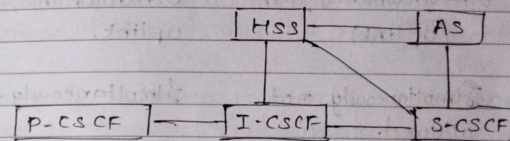


Fig: Reduced IMS Network for VoLTE.

- P-CSCF, Proxy Call State Control funcⁿ: user to network proxy.
- I-CSCF, Interrogating $\rightarrow$: forwarding an initial SIP request to S-CSCF.
- S-CSCF, Serving $\rightarrow$: perform variety of actions within the overall system.
- AS, Applⁿ Server: Handles voice as an applⁿ.
- HSS, Home Subscriber Server: Main subscribed dB; provides details of subscribers to other entities within IMS network.

• Benefits of VoLTE

- i) Provides more efficient use of spectrum than tradⁿ voice.
- ii) Meets the rising demand for richer.
- iii) More reliable services.
- iv) Eliminates the need to hv voice on one network and data on another.
- v) Provides rapid call establishment time.

* Comparison LTE & LTE Adv:

Specifications	LTE	LTE Advanced
Standard	3GPP Release 9	3GPP Release 10
Rollout	2010	2014
Max. no. of users	2, 4 (downlink), 1 in (Uplink)	8 in downlink, 4 in the uplink.
SCH & PUCCH transmission	Simultaneously not allowed.	Simultaneously allowed.
Carrier aggregation	Not supported	Supported
VoLTE	Mobile broadband & VoIP.	Mobile broadband & VoIP.
10 tech. ³⁹	LTE supports 2x2 or 4x4 MIMO.	LTE-A supports upto 8x8 MIMO.
Layer Model	Does not support	Supports
Architecture	Has Simplex Arch.	Has more Complex Arch.
Downlink rate	150 Mbps	1 Gbps
Uplink rate	75 Mbps	500 Mbps

* Self Organising Networks : — (SON)

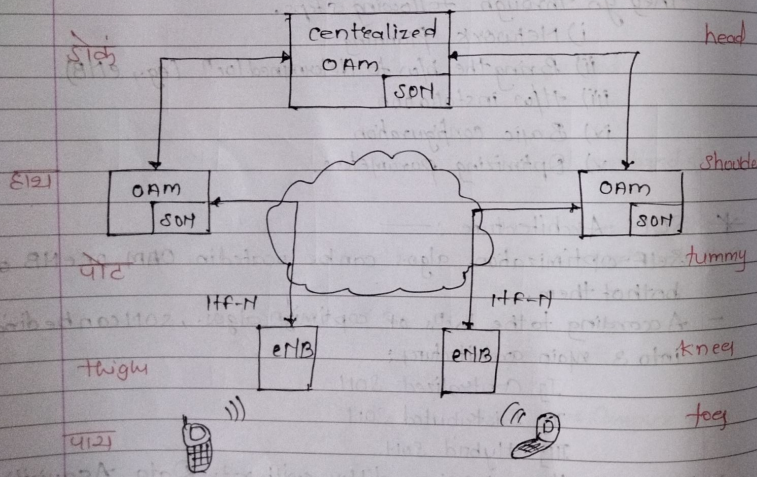
- SON stands for Self Organizing Network.
- It means that just add an eNB wherever you want to put and just connect power and switch on, it would configure all of its configuration by itself and make itself ready for service.
- SON is like a 'Plug-and-Play' functionality.
- The main goal of SON is to automate large portions of human efforts involved in ^{below} mentioned process.
- Normally when a system operator constructs a network, they go through following steps.
 - i) Network Planning
 - ii) Bring the h/w to determined locⁿ (e.g., eNB)
 - iii) H/w installation
 - iv) Basic configuration
 - v) Optimizing parameters.

* SON Architecture : —

- Self-optimization algs can be located in OAM or eNB or both of them.
- According to the locⁿs of optimizn algs., SON can be divided into 3 main architectures:
 - I) Centralized SON
 - II) Distributed SON
 - III) Hybrid SON.
- The all three versions differ with r.t. Data Acquisition, Data Processing and Configuration mgmt.

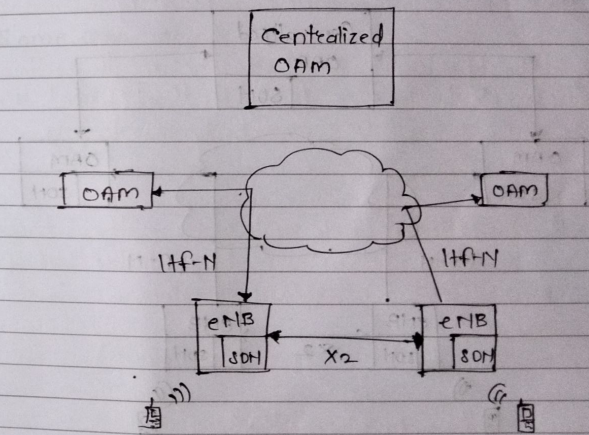
I Centralized SON: —

- In Centralized SON, Optimizⁿ algo^s are stored and executed from the OAM system.
- Here, the SON functionality resides in a small number of locations, at a high level in the architecture.
- Fig. shows an eg. of Centralized SON. Here all SON func^s are located in OAM system, so it's easy to deploy them but does not support those simple quick optimizⁿ req^s.
- To implement Centralized SON, existing Florthbound Interface (Ft-F), which is the interface betⁿ the Element Manager and the Network Manager, needs to be extended.



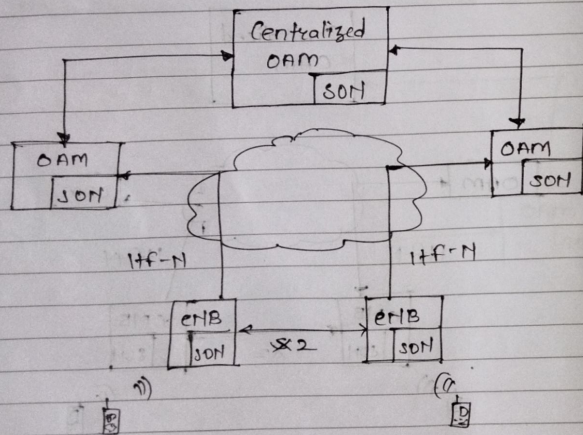
II Distributed SON: —

- In Distributed SON, optimizⁿ algo^s are executed at in eNB.
- SON functionality resides in many locations at a relatively low level in the architecture.
- This increases deployment effort.
- Fig. shows an example of Distributed SON. When this arch. is implemented in large no. of nodes, it has to be ensured that there is a coordination betⁿ them so that the network as a whole is optimized.



III Hybrid SON: —

- In Hybrid SON, part of the optimizⁿ algo are executed in OAM system, while others are executed in eNB.
- fig. shows an eg. of Hybrid SON.
- In Hybrid SON, simple & quick optimizⁿ schemes are implemented in eNB, and complex optimizⁿ schemes are implemented in OAM; so as to provide flexibility to support diff. kind of optimization cases.
- But on other hand, it cost lot of deployment effort and interface extension work.



Module 5

Mobility Management.

- ✓ 1] Explain Mobile IP. UT1
- ✓ 2] Write shortnote: Macro Mobility. UT2
- ✓ 3] Describe use of Cellular IP. May 23, MUSA
- ✓ 4] How IP Mobility is achieved in Wireless Network. May 23
- ✓ 5] Shortnote: IPV6. Dec 23
- ✓ 6] What is need of Micro Mobility? Explain cellular IP in detail. May 24, MUSA
- ✓ 7] Shortnote: IP Mobility. May 25

Some QNO: chp. 3.

* Mobile IP / IP Mobility : —

- IP Mobility is a mechanism that allows a mobile device to move from one network to another network without changing its permanent IP Address.
- ~~Mobile~~ Mobile IP was the first commⁿ protocol developed by IETF to support IP mobility with the current version of IPv4.
- Later Mobile IPv6 (MIPv6) was developed as an update to IPv6 protocols to support IP mobility.
- IP Mobility refers to set of mechanisms that allow an IP mobile node to move freely betⁿ different IP networks while maintaining IP connectivity in a transparent way.
- IPv4 :- If the mobile node moves without changing its address, it loses routing; and if it does change its address, it loses connections.

* IPv6 : — (Internet Protocol version 6)

- To overcome these problems, IPv6 also known as IPng (Internet Protocol next generation) was proposed.
- In IPv6, the Internet Protocol was extensively modified to accommodate the growth and new demands of the Internet.
- The format & length of the IP address were changed along with the packet format.
- The fast spreading use of Internet and new services such as Mobile IP, IP telephony and IP capable mobile

telephony, may require the total replacement of IPv4 by IPv6.

• Advantages :

i) Larger address space -

- An IPv6 address is 128 bit long
- Compared with 32 bit long IPv4 address, this is huge increase in address space.

ii) Better header format -

- IPv6 uses new format for header in which options are separated from the base header and inserted when needed.
- This simplifies and speeds up the routing process.

iii) Allow for extension -

- designed to allow extensⁿ of protocol if req^d by applⁿ

iv) Support for more security -

- Encryptⁿ & authenticatⁿ options in IPv6 provide confidentiality and integrity of the packet.

• features of IPv6 to support mobility :

- i) In every IPv6 node address autoconfiguration mechanism is built.
- ii) Neighbor discovery mechanism is also mandatory for every IPv6 node.
- iii) A soft handover is possible with IPv6.

• Limitation :

It does not solve any firewall or privacy problems.

* Macro Mobility: —

- Macro Mobility refers to the movement of a mobile user across diff. network or large geographical areas, where the user changes from one network domain to another.

- Characteristics:

1. Covers large geographical areas.
2. Involves change of IP network or domain.
3. Req^d "loc" update & rerouting of data.
4. Higher delay compared to Micro Mobility.

- Working:

When a user moves

1. Device detects a new network.
2. Registers with the new network.
3. Updates its "loc" in the system.
4. Ongoing sessions are maintained.

- Protocols used:

- Mobile IP (MIP) → Main protocol for Macro mobility.
- Supports commⁿ across diff. networks.

- Features:

- Maintains connectivity across wide ^{area} ~~across~~
- Enabling roaming
- Handles handover betⁿ networks.

- Adv

- Supports global mobility
- Essential for Cellular System
- Enable seamless commⁿ across networks.

Disadv

- Higher latency during handoff
- Complex mgmt
- Possible packet loss during transition.

* Micro Mobility: —

- Mobile IP represents a simple and scalable global mobility solⁿ but lacks support for the fast handoff control and paging.
- Imagine a large no. of mobile devices changing networks quite frequently; a high load on the "Home Agents" as well as on "Networks" is generated by registration and binding update message.
- IP Micro Mobility protocols can complement Mobile IP by offering fast and almost seamless handover control in limited geographical areas.
- Solⁿ of Micro mobility ⇒ keep the frequent updates generated by local changes at the points of attachments away from the Home network and only inform the home agent about major changes i.e. changes in region.

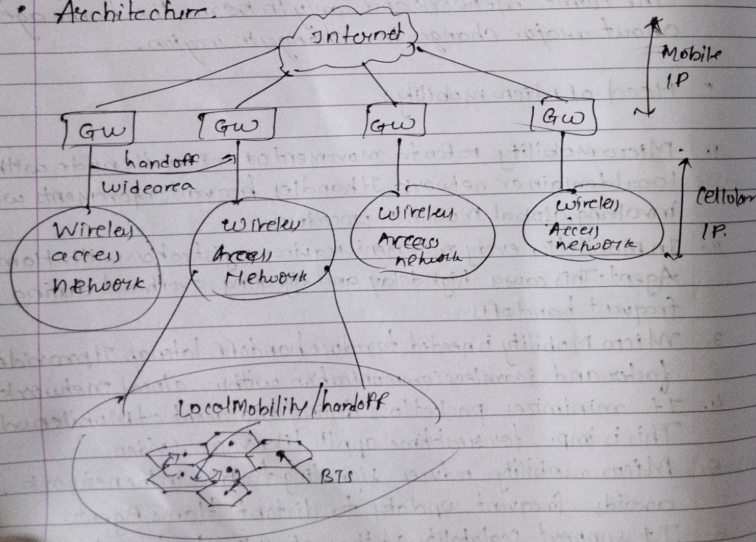
• Need of Micro mobility:

1. Micro Mobility refers to movement of a Mobile node within a local domain or network. It handles frequent movements without involving global Mobile IP procedures.
2. In Mobile IP, every movement requires registration with Home Agent. This causes high delay and signaling overhead during frequent handoffs.
3. Micro Mobility is needed to reduce handoff latency. It provides faster and seamless communication within a local network area.
4. It minimizes packet loss during movement of Mobile nodes. This is imp. for real-time applⁿ like voice & video.
5. Micro mobility reduces signaling traffic in the network. It avoids frequent updates to distant Home Agents.
6. It supports scalability of the network by handling mobility locally.

* Cellular IP: —

- Cellular IP is a Micro Mobility protocol designed to support fast handoff & efficient routing in wireless networks.
- It works within a domain & complements Mobile IP. It avoids frequent communication with Home Agent for local movement.
- Cellular IP integrates routing and mobility mgmt.
- Cellular IP (CIP) is a new robust, simple, and flexible protocol for highly mobile hosts.
- It can accommodate large no. of users by separating idle hosts from active hosts.

• Architecture.



- An imp. component of a Cellular IP network is the Base Station (BS). A Cellular IP network consists of several interconnected BSs.
- The BS communicate with Mobile Host (MH) via wireless interface and also route IP packets inside cellular network.
- The Base Station are built on regular IP forwarding engines, but IP routing is not replaced by Cellular IP routing and Cellular mgmt locⁿ.
- CIP gateway router connects IP network and regular Internet.
- ~~Mobile IP~~
- Mobility betⁿ gateways is managed by Mobile IP while Mobility within access networks is handled by Cellular IP.

• Adv: —

- i) Cheap passive Connectivity.
- ii) flexible handoff.
- iii) Efficient locⁿ mgmt.

• Use of Cellular IP: —

- i) Minimizes delay in packet delivery.
- ii) It is used in cellular networks and wireless Internet systems.
- iii) It is useful in wireless access networks where users move frequently.
- iv) It improves scalability of networks.
- v) Supports paging for idle Mobile nodes.